



Title: Implementation of PL/SQL stored procedure & functions.

Aim: To write and evaluate PL/SQL Stored procedure & function to perform a suitable task on the database & demonstrate their use.

Objective:

1. To understand the concept of PL/SQL procedure & functions.
2. To learn how to create & execute stored procedure.
3. To learn how to create use functions in SQL queries.
4. To differentiate between ~~procedures~~ procedures & function.
5. To apply PL/SQL in real database operations.



## Theory:

### 1) Introduction:

PL/SQL (procedural language/Structured Query language) is Oracle's extension of SQL that allows procedural programming features like loops, conditions & functions.

It helps in writing efficient & reusable database programs.

PL/SQL Supports:

- Stored procedures
- Functions
- Triggers.

### 2) Stored Procedures:

is a named block of PL/SQL code that help performs a specific task & can be stored in a database.

Syntax:

```
CREATE OR REPLACE PROCEDURE  
procedure-name (parameter)  
IS
```



BEGIN

END;

----- PL/SQL statements

Ex:-

CREATE OR REPLACE PROCEDURE get-salary (emp-id Number)

IS

sal NUMBER

BEGIN

SELECT salary INTO sal FROM

employees WHERE id = emp-id;

DBMS-OUTPUT.PUT-LINE ('salary' || sal);

END;

Execution:

BEGIN

get-salary (101);

END;

### 3) Function :

A function is similar to a procedure but always returns a value.



Syntax:

```
CREATE OR REPLACE Function Function-name (parameter)
RETURN datatype
IS
```

```
BEGIN
```

```
    RETURN Value;
```

```
END;
```

Example:

```
CREATE OR REPLACE FUNCTION calculate-bonus
(salary - NUMBER)
```

```
RETURN NUMBER
```

```
IS
```

```
BEGIN
```

```
    RETURN salary * 0.10;
```

```
END;
```

Usage:

```
SELECT calculate bonus (50000) from dual;
```

\* Characteristics of PL/SQL:

- Block structured language
- Supports Variable loops & conditions
- Improve performance by reducing network traffic.



- Error handling using EXCEPTION
- Code reusability through Procedure & Functions.

Diff. bet<sup>n</sup>. Procedure & Function.

| Procedure                                   | Function                             |
|---------------------------------------------|--------------------------------------|
| i) Procedure language does not return value | i) Function must return a value      |
| ii) Called using EXEC or BEGIN --- END      | ii) Usage can be used in SQL queries |
| iii) IN, OUT, IN OUT                        | iii) Only IN                         |
| iv) Perform actions                         | iv) Compute and return value.        |
| v) <u>Ex</u> : Data manipulation            | v) <u>Ex</u> : Calculation.          |



Conclusion: In this practical, we successfully created & executed a PL/SQL stored procedure & function. The procedure was used to perform a database operation while the function returned a calculated value. This demonstrated how PL/SQL enhances database functionality by enabling reusable efficiencies.

| Timely submission<br>(5) | Journal Presentation<br>(5) | Performance<br>(5) | Understanding<br>(5) | Oral<br>(5) | Total<br>(25) | Sign with<br>Date   |
|--------------------------|-----------------------------|--------------------|----------------------|-------------|---------------|---------------------|
| 5                        | 5                           | 4                  | 4                    | 3           | 21            | <u>B</u><br>26/3/20 |